REXUS BRYAN L. GAN

CAD & DESIGN ENGINEERING

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PROFESSIONAL SUMMARY

Aeronautical Engineering graduate (BS) with hands-on experience in 3D CAD modeling, simulation, and engineering design. Proficient in **FreeCAD** and **SolidWorks**, with additional skills in Blender, Python, OpenFOAM, and OpenVSP. Self-taught through project work, with a portfolio that includes a rim-driven propeller model, a complete aircraft design, and an automated airfoil-selection tool. Passionate about teaching parametric modeling and open-source CAD workflows, and about turning complex engineering concepts into clear, practical lessons for students.

TECHNICAL SKILLS

CAD & 3D Modeling: FreeCAD, SolidWorks, Blender; 3D part and assembly modeling

Aeronautical Engineering: Conceptual aircraft/UAV design, weight estimation, airfoil fundamentals, propeller and climb-performance calculations

Simulation & Analysis: OpenFOAM and Salomi exposure; external aerodynamics and CFD fundamentals

Programming & Tools: Python for engineering calculations and automation; Git and GitHub; technical web content

PROJECTS

Conceptual UAV / Aircraft Design

- Developed conceptual design work involving configuration choices, component placement, and preliminary weight estimation.
- Made Software that would streamline the Conceptual Design Pipeline to allow rapid calculation of weight estimation at varying mission profiles, Parametric Modeling for Wings and Fuselage,
- Applied engineering assumptions and first-principles calculations to support early-stage design decisions.

Airfoil and Weight-Distribution Calculations

- Explored airfoil-related engineering calculations and weight-distribution methods; used scripting to support repeatable analysis and optimization tasks.

Propeller and Climb-Performance Modeling

- Worked through propeller and aircraft performance relationships to evaluate climb and flight-performance concepts.

Engineering Learning Platform & Portfolio

- Built a web-based engineering learning project with a question-bank structure and interactive learning modes; published work through GitHub Pages.

FreeCAD 3D Parametric Modeling

SolidWorks 3D Parametric Modeling and Fluid Simulation

EDUCATION

Mats College of Technology | Bachelor of Science in Aeronautical Engineering

Graduation year: 2026

AWARDS & RECOGNITION

- Best in Aircraft Design
- Best in Feasibility Study
- Best in UAV Design

AFFILIATIONS

- **SAEP SCF -Society of Aerospace Engineering of the Philippines (2021-2022)** | Education and Research Director
- **Mindanao Aeronautical Engineering Student Organization (MAESO**) (2021-2022) | 1st Year Representative
- **Mindanao Aeronautical Engineering Student Organization (MAESO** (2022-2023) | 2nd Year Representative
- **Mindanao Aeronautical Engineering Student Organization (MAESO** (2024-2026) | Education and Research Director
- **Society of Engineering and Technology (2025-2026**) | Info media

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REFERENCE

ENGR. Calvin Jhon Labial | Aeronautical Engineer

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